Autoregressive Models

Set up

```
library(fpp3)
library(tidyverse)
library(slider)
library(gridExtra)
```

1. Residual analysis

a. What is the difference between innovation residuals and regular residuals? Why do we perform residual analysis on the former rather than the latter?
b. Starting with the following code snippet, fit several models to the Australian takeaway turnover time series and analyze their residuals. Which models have a good fit? Compare this with their CV error.

```
takeaway <- aus_retail |>
  filter(Industry == "Takeaway food services") |>
  summarise(Turnover = sum(Turnover))
```

If a transformation is performed, the innovation residuals are the one-step ahead forecasting errors, $y_t - \hat{y}_{t|t-1}$ for the transformed time series. The regular residuals are the errors for the original time series $x_t - \hat{x}_{t|t-1}$.

We perform residual analysis for the former because the modeling assumption is that the innovation residuals are white noise.

```
fit <- takeaway |>
  model(
    naive = NAIVE(Turnover),
    drift = NAIVE(Turnover ~ drift()),
    mean = MEAN(Turnover),
    snaive = SNAIVE(Turnover),
    snaive_drift = SNAIVE(Turnover ~ drift()),
    ets = ETS(Turnover),
    ets_log = ETS(log(Turnover))
  )

fit |>
```

```
  augment() |>
  features(.innov, ljung_box, lag = 24) |>
  arrange(lb_stat)
```

```
# A tibble: 7 × 3
  .model       lb_stat lb_pvalue
  <chr>          <dbl>     <dbl>
1 ets             89.8  1.57e- 9
2 ets_log         93.9  3.20e-10
3 naive         1007.   0
4 drift         1007.   0
5 snaive_drift  1620.   0
6 snaive        1620.   0
7 mean          8738.   0
```

```
fit
```

```
# A mable: 1 x 7
    naive         drift    mean   snaive      snaive_drift
ets
  <model>       <model> <model>  <model>           <model>
<model>
1 <NAIVE> <RW w/ drift>  <MEAN> <SNAIVE> <SNAIVE w/ drift>
<ETS(M,Ad,M)>
# ℹ 1 more variable: ets_log <model>
```

```
takeaway |>
  stretch_tsibble(.step = 20, .init = 200) |>
  model(
    naive = NAIVE(Turnover),
    drift = NAIVE(Turnover ~ drift()),
    mean = MEAN(Turnover),
    snaive = SNAIVE(Turnover),
    snaive_drift = SNAIVE(Turnover ~ drift()),
    ets = ETS(Turnover ~ error("M") + trend("Ad") + season("M")
    ets_log = ETS(log(Turnover) ~ error("A") + trend("A") + sea
  ) |>
  forecast(h = 6) |>
  accuracy(takeaway) |>
  select(.model, RMSSE, MASE) |>
  arrange(MASE)
```

```
Warning: The future dataset is incomplete, incomplete out-of-
sample data will be treated as missing.
5 observations are missing between 2019 Jan and 2019 May
```

```
# A tibble: 7 × 3
  .model       RMSSE   MASE
  <chr>        <dbl>  <dbl>
1 ets_log      0.740  0.707
2 ets          0.781  0.773
3 snaive_drift 0.769  0.802
4 drift        0.888  0.836
5 naive        0.910  0.851
6 snaive       1.08   1.21
7 mean         9.30  10.6
```

The ranking via the LB test statistic and the CV error do not entirely match. Note that the LB statistic is a reflection of training error and measures "goodness of fit". Furthermore, the residuals from different methods may be on different scales and hence are not entirely comparable.

2. ACF for AR(2)

a. Using `ARMAacf`, plot the ACF for AR(2) models with the following coefficients:

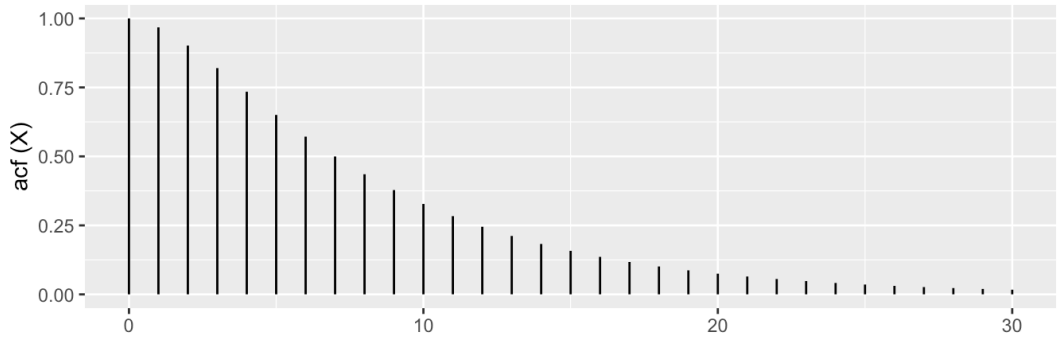
- (X_t): $\phi_1 = 1.5$, $\phi_2 = -0.55$
- (Y_t): $\phi_1 = 1.5$, $\phi_2 = -0.75$

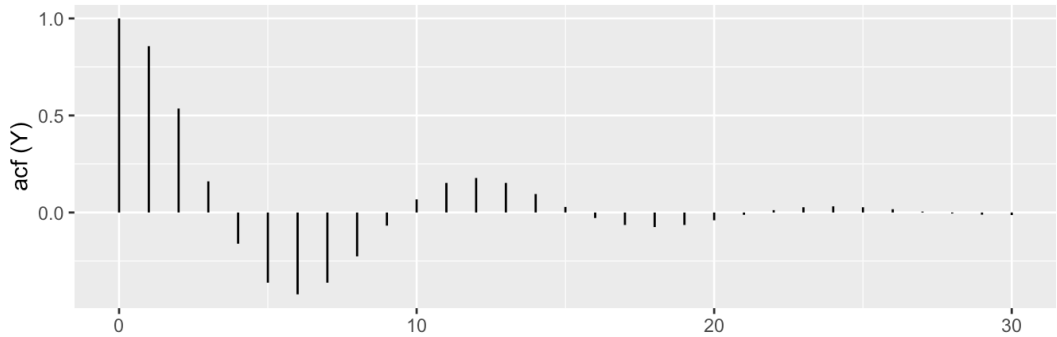
b. What is the qualitative behavior of the ACFs?
c. From the ACF plots, try to guess what patterns you may observe from the sample trajectories.

```r
acf_dat <- tibble(
    lag = 0:30,
    X_acf = ARMAacf(ar = c(1.5, -0.55), lag.max = 30),
    Y_acf = ARMAacf(ar = c(1.5, -0.75), lag.max = 30))

plt1 <- acf_dat |>
    ggplot() + geom_linerange(aes(x = lag, ymax = X_acf, ymin =
    xlab("") + ylab("acf (X)")
plt2 <- acf_dat |>
    ggplot() + geom_linerange(aes(x = lag, ymax = Y_acf, ymin =
    xlab("") + ylab("acf (Y)")

grid.arrange(plt1, plt2, nrow = 2)
```





---

3. Sample trajectories

a. Using `arima.sim()`, draw sample trajectories from AR(2) models with different coefficients.
b. Fit AR(2) models to these generated time series using `ARIMA()`.
c. What are the fitted coefficients?
d. Plot the forecast curves.

```
set.seed(5209)
n <- 80
h <- 20
ar2_data <-
    tibble(t = 1:n,
           wn = rnorm(n),
           X = arima.sim(model = list(ar = c(1.5, -0.55)),
                         n = n, innov = wn),
           Y = arima.sim(model = list(ar = c(1.5, -0.75)),
                         n = n, innov = wn)
    ) |>
    as_tsibble(index = t)
```

```
X_mod <- ar2_data |>
  model(X = AR(X ~ order(2)))
```

```r
Y_mod <- ar2_data |>
  model(Y = AR(Y ~ order(2)))

X_mod |> tidy()
```

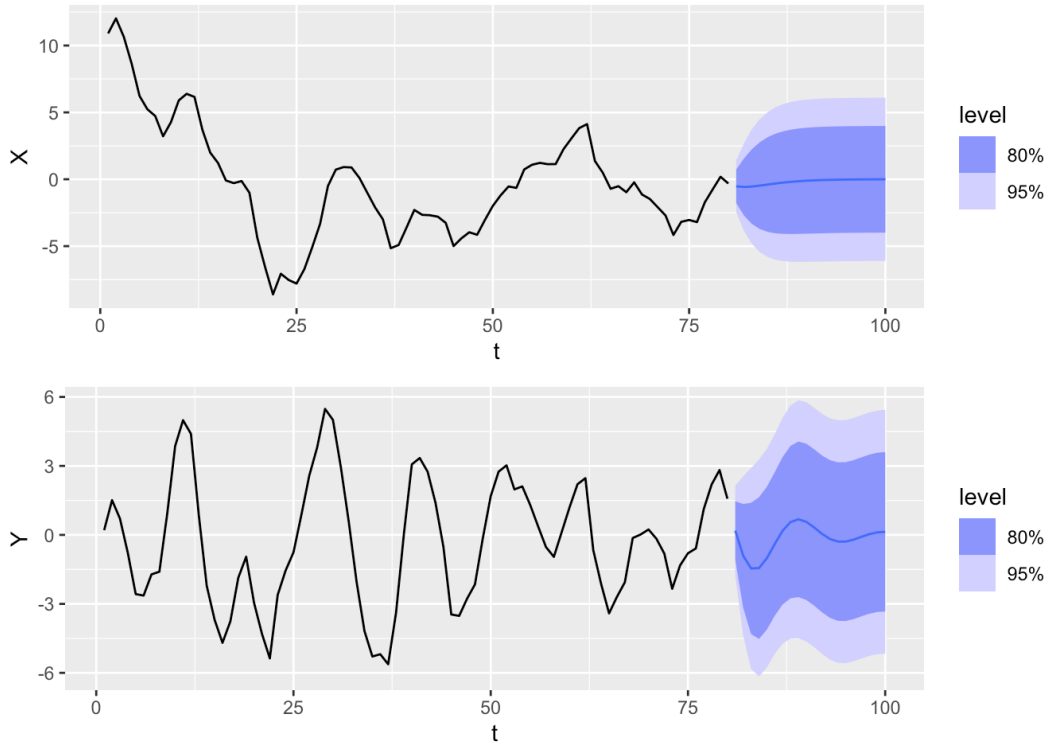
```
# A tibble: 2 × 6
  .model term  estimate std.error statistic  p.value
  <chr>  <chr>    <dbl>     <dbl>     <dbl>    <dbl>
1 X      ar1      1.38     0.0952     14.5  6.92e-24
2 X      ar2     -0.482    0.0911     -5.28 1.11e- 6
```

```r
Y_mod |> tidy()
```

```
# A tibble: 2 × 6
  .model term  estimate std.error statistic  p.value
  <chr>  <chr>    <dbl>     <dbl>     <dbl>    <dbl>
1 Y      ar1      1.46     0.0754     19.3  1.93e-31
2 Y      ar2     -0.747    0.0759     -9.84 2.56e-15
```

```r
plt1 <- X_mod |>
    forecast(h = h) |>
    autoplot(ar2_data) + ylab("X")
plt2 <- Y_mod |>
    forecast(h = h) |>
    autoplot(ar2_data) + ylab("Y")

grid.arrange(plt1, plt2, nrow = 2)
```



4. AR(2) solution

Consider the AR(2) equation

$$X_t = 1.5X_{t-1} - 0.75X_{t-2} + W_t.$$

a. What is the autoregressive polynomial?

b. What is the formula for the stationary solution?